

Visual acuity after Gamma-Knife radiosurgery of choroidal melanomas

Werner Wackernagel,¹ Etienne Holl,² Lisa Tarmann,¹ Alexander Avian,³ Mona Regina Schneider,¹ Karin Kapp,⁴ Gerald Langmann¹

¹Department of Ophthalmology, Medical University Graz, Graz, Austria

²Department of Neurosurgery, Medical University Graz, Graz, Austria

³Institute for Medical Informatics, Statistics and Documentation, Medical University Graz, Graz, Austria

⁴Department of Therapeutic Radiology and Oncology, Medical University Graz, Graz, Austria

Correspondence to

Dr Werner Wackernagel, Department of Ophthalmology, Medical University Graz, Auenbruggerplatz, Graz 4A-8036, Austria; werner.wackernagel@medunigraz.at

Received 21 July 2012

Revised 6 October 2012

Accepted 9 October 2012

Published Online First

7 November 2012

ABSTRACT

Background/aims To report on conservation of visual acuity after Gamma-Knife radiosurgery of choroidal melanoma.

Methods A total of 189 patients with choroidal melanoma were treated with Gamma-Knife stereotactic single-fraction radiosurgery at a single institution between June 1992 and May 2010. The main outcome measure of our retrospective analysis was conservation of pretreatment visual acuity of 20/40 or better, 20/200 or better and counting fingers (CF) or better, over time of follow-up. Patient, tumour and treatment parameters were evaluated as potential risk factors for visual loss.

Results Five years after treatment, the actuarial probability of keeping visual acuity better than 20/40, 20/200 and CF was 13%, 14% and 36%, respectively. The majority of patients (84.7%) encountered a deterioration of vision after treatment. The most important risk factors for visual loss were tumour height, longest basal diameter, distance to the optic disk and/or foveola, and retinal detachment before treatment. Treatment dose, and patient characteristics (age, sex, concurrent systemic diseases) were less important. Local tumour control rate was 94.4% after a median follow-up of 39.5 months.

Conclusions Visual outcome after single-fraction Gamma-Knife radiotherapy is comparable with linear accelerator (LINAC) based fractionated stereotactic radiotherapy, inferior to proton beam radiotherapy, and depends primarily on tumour size, location and pre-existing retinal detachment.

INTRODUCTION

Choroidal melanoma is a rare, but life-threatening disease. The annual incidence of choroidal melanoma is 4–7 cases per million per year, and has remained stable over the past decades.^{1–2} The 15-years melanoma-related mortality is 40–50%, mainly due to metastasis to the liver.³ Over the past decades, eye-sparing treatments have replaced enucleation as initial therapy for small and medium-sized choroidal melanomas in most cases. Plaque brachytherapy and proton beam radiotherapy are the most commonly used treatment modalities. Stereotactic radiotherapy using photons and local resection (trans-scleral and transretinal) are used less frequently by only few ophthalmologists, and still considered investigational. Transpupillary thermotherapy, as a sole treatment, has largely been abandoned, however, it is still being used as an adjunct treatment to radiotherapy. All eye-sparing treatment modalities aim at destroying the

melanoma, retaining the globe, and preserving some useful vision.

Gamma-Knife stereotactic radiosurgery was first used in the field of neurosurgery for the treatment of intracranial lesions, difficult to treat surgically.⁴ It was adopted by some ocular oncologists for the treatment of posterior uveal melanoma, mainly as an alternative to enucleation, when no other globe-preserving treatment options were available or feasible.^{5–13} We have been using radiosurgery for the treatment of choroidal melanoma in selected cases since 1992.⁷ In previous studies, we described the technique we use, tumour regression after treatment, dose distribution and radiation side effects.^{7 14 15} In this study, we report on visual outcome in patients with choroidal melanoma after Gamma-Knife radiosurgery (GKR), and identify risk factors for visual loss after treatment.

METHODS

Patients and data collection

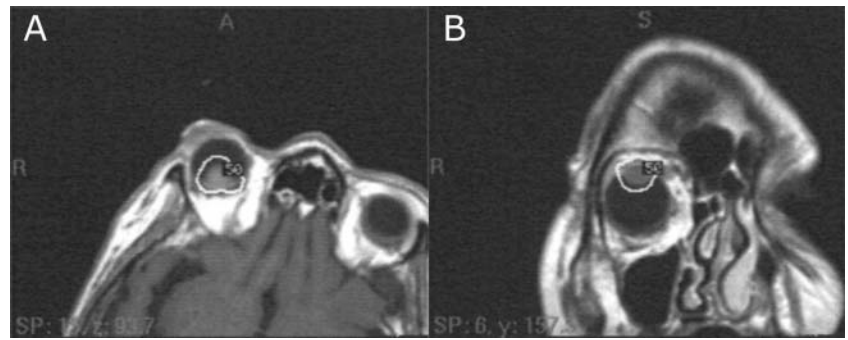
Institutional review board (IRB) approval was obtained to review medical records and radiosurgery treatment plans of patients with choroidal melanoma treated with GKR at University Hospital Graz, between June 1992 and May 2010. Patient data, clinical and treatment parameters have prospectively been recorded in electronic databases at the Department of Neurosurgery (Q&A Database, V4; Symantec Inc), and at the Department of Ophthalmology (Filemaker Pro V3 to 6, Apple Inc). The medical records of all patients were reviewed, and patient data, tumour features, treatment parameters and visual outcome were recorded.

Variables recorded and evaluated for the present study to describe the patient demographics were patient age (years), sex (female or male) and systemic diseases (diabetes mellitus, hypertension, coronary artery disease, other cancer). Variables for ocular status were best-corrected Snellen visual acuity of the affected and of the fellow eye (20/20, 20/25, 20/30, 20/40, 20/50, 20/60, 20/70, 20/80, 20/100, 20/125, 20/160, 20/200, 20/400, counting fingers (CF), hand motions, light perception, no light perception), concurrent abnormalities of the affected and the fellow eye (cataract, pseudo-aphakia, retinal detachment, glaucoma, age-related macular degeneration, diabetic retinopathy, amblyopia).

Variables describing the tumour were height (millimetres, measured by standardised echography), longest basal diameter (millimetres, measured by ultrasound and indirect ophthalmoscopy), minimal distance of the posterior tumour margin

To cite: Wackernagel W, Holl E, Tarmann L, et al. *Br J Ophthalmol* 2013;**97**:153–158.

Figure 1 Gamma-Knife radiosurgery treatment planning. Gamma-Knife treatment plan, showing magnetic resonance images (T1, axial section, A; T1, coronal section, B) of a choroidal melanoma of the right eye. The 50% isodose encompasses the planning target volume.



to the optic disc and foveola (millimetres, based on binocular indirect ophthalmoscopy, fundus photographs and fluorescein-/indocyanin green angiography), anterior tumour margin (ciliary body, anterior or posterior to equator), coronal location (temporal, midline, nasal), sagittal location (superior, horizontal, inferior), tumour shape (dome, mushroom) and pigmentation (melanotic, amelanotic). Recorded treatment parameters were treatment dose (Gray (Gy)) and isodose (percentage of maximum dose) to the planning target volume (PTV). Patients were generally re-examined 1 month after treatment, then every 3 months for the first 2 years after treatment, every 6 months up to 5 years after treatment and once a year thereafter.

Gamma-Knife radiosurgery protocol

GKR was performed as described previously.⁷ In brief, a mixture of 2.5 ml lidocainhydrochloride 2%, 2.5 ml ropivacainhydrochloride 10 mg/ml, and hyaluronidase 150 IE, was given as a retrobulbar injection to immobilise the eye. The stereotactic frame was then fixated to the skull. The 4 rectus muscles were fixated to the stereotactic frame with transconjunctival sutures to ensure immobilisation of the eye throughout treatment. The eye remained immobilised throughout treatment. Magnetic resonance imaging was performed for treatment planning (Leksell GammaPlan software; Elekta, Stockholm, Sweden). Gross volume of the melanoma was identified and a margin of 1 mm was added to create the PTV. A conformation technique was used for treatment planning. Plans were designed to encompass the entire PTV within the 50% isodose (range: 32–80%), and to spare critical intraocular structures only if possible, without compromising coverage of the PTV (figure 1). A detailed dose-volume histogram analysis, and an analysis of dose to the tumour and critical structures has been published.¹⁵

Statistical methods

Data are presented as median and range for continuous data, and absolute and relative frequency for categorical data, respectively. Survival was calculated from the date of GKR until visual loss. Kaplan–Meier estimates and life-table methods were used to describe the risk of visual loss over time. The midpoint estimator was used for interval-censored data, and time to visual loss was defined as the midpoint between the last exam the patient had good visual acuity and the first exam when visual loss was detected. Log-rank test of equality across strata and Cox's univariate proportional hazard model were used to assess the impact of baseline variables on the risk of visual loss. HRs were calculated for each variable. A stepwise multivariate Cox regression analysis, including those variables that showed a *p* value of ≤ 0.05 in the univariate procedure, was used to identify baseline variables as independent risk factors for visual deterioration. For all *p* values, statistical significance level was defined as

0.05. Data were analysed using the statistical software SPSS 19 for Windows (SPSS Inc., Chicago IL, USA).

RESULTS

Description of patient population and tumour characteristics

In total, 189 patients with choroidal melanoma were treated with Gamma-Knife stereotactic radiosurgery between June 1992 and May 2010. Six patients were excluded from the study because of incomplete data, and another six patients were excluded because of previous treatment with TTT (3 patients) or Ruthenium-106 plaque brachytherapy (3 patients). The median follow-up for the 177 patients included in the study was 39.5 months (interquartile range, IQR: 20.6–70.5 months). Patient, tumour and treatment data are detailed in table 1. The tumour was touching the optic disc margin in 39 patients (20.9%) and involving the foveola in 48 patients (27.1%). Visual acuity in the fellow eye was $\geq 20/40$ in 165 patients (93.2%), 20/200–20/50 in 9 patients (5.1%), and $< 20/200$ in 3 patients (1.7%).

Local tumour control, complications and secondary enucleation

Ten patients (5.6%) showed tumour recurrence after a median time of 13.6 months (range: 3.1–60.7 months) after treatment. Sixty-one patients (34.5%) showed an increasing lens opacification after treatment. Thirty-five patients (19.8%) developed rubeosis iridis, 31 patients (17.5%) developed glaucoma. Retinal detachment developed or increased in 73 patients (41.2%) after treatment. Radiation maculopathy was found in 112 patients (63.3%), radiation retinopathy in 116 patients (65.5%), optic nerve neuropathy in 77 patients (43.5%). Tumour vasculopathy was found in 97 cases (54.8%), and vitreous haemorrhage occurred in 35 cases (19.8%). In 25 patients (14.1%) secondary enucleation was required because of tumour recurrence or post-treatment complications.

Visual acuity

Conservation of visual acuity of 20/40 or better

Before treatment, 58 patients (32.8%) had visual acuity of 20/40 or better on the affected eye. The actuarial probability of conservation of 20/40 or better was 0.53 (95% CI 0.40 to 0.66) after 1 year, 0.25 (95% CI 0.14 to 0.37) after 2 years, and 0.13 (95% CI 0.03 to 0.23) after 5 years.

Cox univariate analysis and log-rank test showed increased tumour height ($p < 0.001$), distance from disc or foveola of ≤ 1.5 mm ($p = 0.013$), tumour height of ≥ 5.6 mm ($p = 0.002$), and pre-existing retinal detachment ($p = 0.010$) to be significant risk factors for visual loss (table 2).

In 41 of the 58 patients (70.7%) with good initial visual acuity, the tumour was within a distance of ≤ 1.5 mm from the

Table 1 Summary of patient and tumour characteristics (n=177)

Characteristic	Value (%)
Sex	
Male	91 (51.41)
Female	86 (48.59)
Laterality	
Right	83 (46.89)
Left	94 (53.11)
Age (years)	62.1 (median; IQR: 51.8–70.5)
Follow-up (months)	39.5 (median; IQR: 20.6–70.5)
Diabetes	22 (12.4)
Systemic hypertension	62 (35.0)
Coronary artery disease	25 (14.1)
Other cancer	12 (6.8)
Tumour height (mm)	6.1 (median; IQR: 4.4–8.2)
<5.6 mm	76
≥5.6 mm	101
Longest basal diameter (mm)	11.7 (median; IQR: 9.4–14.3)
TNM	
T1	28
T2	74
T3	71
T4	4
Shape	
Dome	135 (76.3)
Mushroom	42 (23.7)
Retinal detachment	
Present	108 (61.0)
Absent	69 (39.0)
Tumour pigmentation	
Melanotic	165 (93.2)
Amelanotic	12 (6.8)
Coronal location	
Nasal	28 (15.8)
Midline	59 (33.33)
Temporal	90 (50.85)
Sagittal location	
Superior	49 (27.7)
Horizontal	80 (45.2)
Inferior	48 (27.1)
Anterior margin	
Ciliary body	7 (4.0)
Anterior to equator	49 (27.7)
Posterior to equator	121 (68.4)
Posterior margin	
≤1.5 mm from optic disk or fovea	124 (70.1)
1.6–3.0 mm from optic disk or fovea	24 (13.6)
≥3.0 mm from disk or fovea	29 (13.4)
Treatment dose	
25 Gy	24 (13.6)
30 Gy	87 (49.2)
35 Gy	12 (6.8)
40 Gy	12 (6.8)
40–80 Gy	42 (23.7)

optic disc or foveola. Their relative risk of losing good visual acuity was 2.5 times the risk of patients with tumours farther than 1.5 mm from the optic disk or foveola. The HR per millimetre increase in distance to disk/foveola was 0.73 (CI 0.57 to 0.92). Figure 2A and 2B show a comparison of Kaplan–Meier estimates for keeping visual acuity of ≥20/40 according to

Table 2 Loss of visual acuity of 20/40 or better. Univariate analysis

Characteristic	Number at risk	Hazard ratio (95% CI)	p Value
Sex			
Male	27	1	0.416
Female	31	1.27 (0.71 to 2.27)	
Age			
Per-year increase		0.99 (0.97 to 1.02)	0.583
Posterior margin			
≥1.5 mm from disk/foveola	17	1	0.013
≤1.5 mm from disk/foveola	41	2.45 (1.18 to 5.10)	
Tumour height			
<5.6 mm	31	1	0.002
≥5.6 mm	27	2.46 (1.36 to 4.42)	
Retinal detachment			
Absent	28	1	0.010
Present	30	2.19 (1.18 to 4.06)	

posterior tumour extension and tumour height. Treatment dose, age, gender and systemic diseases (diabetes, systemic hypertension, coronary artery disease) were not significant risk factors. The multivariate Cox analysis identified tumour height, distance from optic disk or foveola, and retinal detachment as independent risk factors for visual deterioration (table 5).

Conservation of visual acuity of 20/200 or better

There were 128 patients (72.3%) with initial visual acuity of 20/200 or better on the affected eye. The actuarial probability of keeping visual acuity of 20/200, was 0.64 (95% CI 0.55 to 0.72) 1 year after treatment, 0.41 (95% CI 0.32 to 0.50) after 2 years, and 0.13 (95% CI 0.05 to 0.22) after 5 years. Log-rank test and univariate Cox analysis revealed tumour height of ≥5.6 mm ($p=0.004$), longest basal diameter ($p=0.001$), distance to disc or foveola ≤1.5 mm ($p=0.027$), initial visual acuity ($p=0.001$) and retinal detachment prior to treatment ($p<0.001$) as significant risk factors for loss of vision of 20/200 (table 3).

In 84 patients (65.6%), the melanoma was within 1.5 mm from the optic disk or foveola, and those patients had a 1.7 times higher risk for deterioration of vision (figure 2C). In the multivariate Cox regression model, tumour height, distance to optic disk or foveola, retinal detachment before treatment and initial (pretreatment) visual acuity were significant (table 5).

Conservation of visual acuity of CF or better

We identified 156 patients (88.1%) with pretreatment visual acuity of CF or better. One year after treatment, the probability of keeping this visual acuity was 0.72 (95% CI 0.65 to 0.80). Two-year and 5-year probabilities were 0.57 (95% CI 0.49 to 0.65) and 0.37 (95% CI 0.27 to 0.46), respectively. In the univariate analysis, tumour height of ≥5.6 mm ($p<0.001$), initial visual acuity ($p<0.001$), retinal detachment ($p<0.001$), nasal location ($p=0.003$) and tumour shape ($p=0.037$) were the significant risk factors (table 4).

As for keeping a visual acuity of 20/40 and 20/200, the other parameters analysed (patient age, sex, systemic diseases) were not significant. There was a trend for higher treatment doses ($p=0.055$), cataract of the affected eye ($p=0.057$) and longer basal diameter ($p=0.065$) to result in a higher probability of

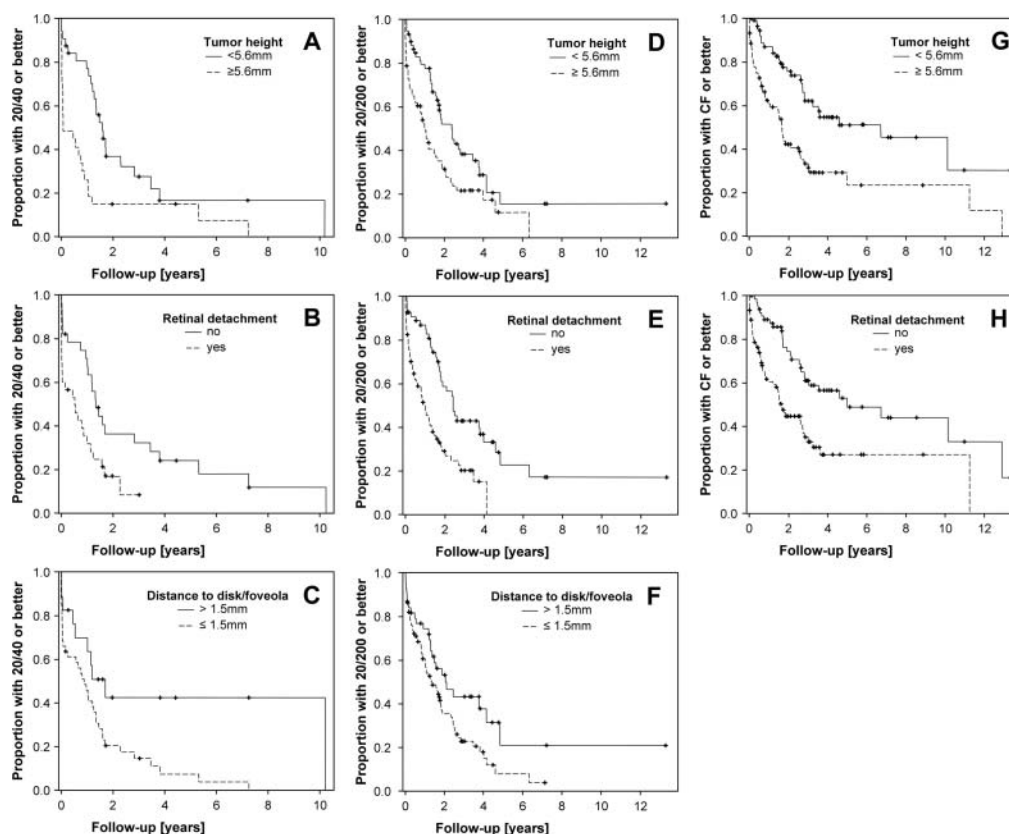


Figure 2 Kaplan–meier curves for keeping pretreatment visual acuity. Probability of preservation of $\geq 20/40$ (A–C), $\geq 20/200$ (D, E, F) and counting fingers (G, H) after treatment, according to tumour height (A, D, G), pre-existing retinal detachment (B, E, H) and posterior tumour extension (C, F). For visualisation of the continuous variable tumour height a threshold of 5.6 mm was chosen.

visual loss, which, however, were not quite significant. In Cox multivariate analysis, tumour height (≥ 5.6 mm), retinal detachment and initial visual acuity remained significant (table 5).

Final visual acuity

Deterioration of visual acuity occurred in 150 patients (84.7%). In 26 patients (15.3%), visual acuity remained stable (15) or

Table 3 Loss of visual acuity of 20/200 or better

Characteristic	Number at risk	HR (95% CI)	p Value
Sex			
Male	65	1	0.694
Female	63	1.09 (0.72 to 1.65)	
Age			
Per-year increase		1.00 (0.99 to 1.02)	0.730
Initial visual acuity			
Per Snellen line worse acuity		1.15 (1.06 to 1.25)	0.001
Posterior margin			
≥ 1.5 mm from disk/foveola	44	1	0.027
≤ 1.5 mm from disk/foveola	84	1.68 (1.06 to 2.68)	
Tumour height			
< 5.6 mm	61	1	0.004
≥ 5.6 mm	67	1.85 (1.21 to 2.83)	
Tumour longest basal diameter			
Per mm increase		1.13 (1.05 to 1.22)	0.001
Tumour shape			
Dome	102	1	0.001
Mushroom	26	2.57 (1.10 to 6.00)	
Retinal detachment			
Absent	54	1	< 0.001
Present	74	2.52 (1.60 to 3.98)	

Univariate analysis.

Table 4 Loss of visual acuity of counting fingers or better

Characteristic	Number at risk	Hazard ratio (95% CI)	p Value
Sex			
Male	81	1	0.109
Female	75	1.42 (0.92 to 2.20)	
Age			
Per-year increase		1.01 (0.99 to 1.03)	0.231
Initial visual acuity			
Per Snellen line worse acuity		1.08 (1.05 to 1.12)	< 0.001
Tumour height			
< 5.6 mm	71	1	< 0.001
≥ 5.6 mm	85	2.45 (1.55 to 3.86)	
Coronal location			
Midline	48	1	0.003
Nasal	25	2.60 (1.40 to 4.85)	0.762
Temporal	83	1.08 (0.65 to 1.811)	
Tumour shape			
Dome	102	1	0.037
Mushroom	26	1.67 (1.03 to 2.70)	
Retinal detachment			
Absent	66	1	< 0.001
Present	90	2.39 (1.50 to 3.80)	

Univariate analysis.

Table 5 Conservation of visual acuity

Visual acuity	Characteristic	Hazard ratio (95% CI)	p Value
20/40 or better	Tumour height ≥ 5.6 mm	2.75 (1.49 to 5.08)	0.001
	Retinal detachment	2.25 (1.20 to 4.21)	0.011
	Distance to disk/foveola ≤ 1.5 mm	2.67 (1.28 to 5.57)	0.009
20/200 or better	Tumour height ≥ 5.6 mm	1.82 (1.17 to 2.83)	0.008
	Retinal detachment	2.19 (1.36 to 3.54)	0.001
	Distance to disk/foveola ≤ 1.5 mm	2.47 (1.50 to 4.07)	<0.001
	Initial visual acuity	4.35 (1.83 to 10.33)	0.001
Counting fingers or better	Tumour height ≥ 5.6 mm	1.91 (1.17 to 3.12)	0.009
	Retinal detachment	1.876 (1.07 to 2.89)	0.025
	Initial visual acuity	2.17 (1.54 to 3.04)	<0.001
Multivariate analysis.			

improved (12). At the end of follow-up, 11 patients (6.2%) had visual acuity of 20/40 or better, compared with 58 patients (32.8%) before treatment. Forty-one patients (23.2%) had a final visual acuity of 20/200 or better (before treatment: 128 patients, 72.3%), and 73 (41.2%) had CF or better (before treatment: 156 patients, 88.1%). Fifty-six patients (33.3%) finally had only light perception or could perceive hand motions, and 48 patients (26.0%), including 25 patients with secondary enucleation, had no light perception on the treated eye.

DISCUSSION

Globe-preserving treatment today is the treatment of choice for most small and medium-sized uveal melanomas.² We have been using GKR since 1992, as an alternative to enucleation in patients with uveal melanoma unsuitable for plaque brachytherapy or surgery. Proton beam therapy is not available in Austria till today. Details on the treatment technique we used, tumour regression after treatment, the effect of treatment dose and dose to critical structures on outcome, and the incidence of radiation retinopathy have been published.^{7 14 16 17} In the present study, we report on conservation of vision after GKR for a large number of patients, over a long follow-up time, analysing data which were prospectively recorded in electronic databases.

The most important clinical risk factors for visual loss were distance of the tumour from the optic disk or fovea, tumour height, longest basal diameter, retinal detachment before treatment and initial visual acuity. These parameters are well-recognised risk factors, repeatedly reported for proton beam radiotherapy, and also for plaque brachytherapy.^{18–21} The optic nerve and the fovea are most sensitive to irreversible damage from radiation. Tumour extension close to those high-risk structures inevitably results in high doses of radiation to those structures, and direct radiation-related complications (radiation optic neuropathy and maculopathy). They are also more likely to be indirectly affected, by bleeding and exudation from a tumour close to the disk or fovea after radiation. A large tumour and treatment volume has been identified as a major risk factor for radiogenic side effects after GKR due to tumour necrosis and inflammation.¹⁵ Large tumour size and proximity of the posterior tumour margin to the fovea are also associated with reduced initial visual acuity, because of the tumour or retinal detachment affecting the fovea. Therefore, the identified risk factors appear plausible. Large tumour size is also a risk factor for retinal detachment before treatment, and we thought about excluding retinal detachment as a separate risk factor from our analysis.²² However, retinal detachment is also associated with distinct histological features of choroidal melanoma, especially

microvascular factors, which might be important for the reaction of the melanoma to radiotherapy and for visual outcome.²² We therefore included retinal detachment in our multivariate analysis.

By contrast with other studies, patient age, systemic hypertension, diabetes and coronary artery disease were not significant risk factors for visual loss in our study.²⁰ There was a trend towards unfavourable visual outcome for higher treatment doses, which however, failed to reach the predefined p-level for statistical significance. Larger numbers of patients might be required to identify less important risk factors. Direct comparisons with visual outcome published for other treatment modalities are tempting, however difficult and sometimes misleading. Most of our patients (83.6%) had tumours within 3 mm from the optic disk or foveola, which is comparable with data published by Dunavoelgyi *et al*²³ on LINAC-based stereotactic radiotherapy. Tumour height and longest diameter are also similar. Visual outcome in our study is almost identical to their report. Eighty-three percent of their patients had visual acuity $\geq 20/200$ before treatment, and 8.8% 5 years after treatment (vs 85.3% and 12.0% in our group). A study by Seddon *et al*¹⁹ on results after proton beam radiotherapy also presented comparable data. Their results suggest a higher probability of retaining 20/200 compared with our study. Damato *et al*²¹ reported comparably favourable visual results after proton beam radiotherapy. However, tumour size (median height 3.0 mm) and tumour location were substantially different, and a direct comparison is not possible. A study on visual acuity after proton beam treatment of extra-large uveal melanomas shows less favourable results.²⁴

GKR is attractive in many ways. It is minimally invasive and does not require preparatory surgery before treatment. The treatment takes only a few hours and can be done as an outpatient procedure. The steep dose cut-off, and the radiobiological characteristics, are well suited for the treatment of choroidal melanomas.²⁵ Visual outcome is comparable with results after LINAC-based stereotactic radiotherapy. Proton beam therapy is more easily accessible today than it was 20 years ago when we started GKR. A matched-pair study, comparing radiosurgery against proton beam therapy is necessary to show if there is a possible indication for GKR in the treatment of uveal melanoma in the future.

Contributors All authors substantially contributed to preparation of the manuscript (according to ICMJE requirements). In detail, their authorship is based on contribution. WW: conception and design of the study, data acquisition, analysis and interpretation of data, drafting and revising the manuscript. EH, LT and MS: data acquisition, drafting and revising the manuscript. AA: conception and design of the study, analysis and interpretation of data, and revising the manuscript. KK and GL: conception and design of the study, interpreting the results and revising the manuscript.

Funding This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests All authors have completed the Unified Competing Interest form at http://www.icmje.org/coi_disclosure.pdf (available on request from the corresponding author) and declare: no support from any organisation for the submitted work, no financial relationships with any organisations that might have an interest in the submitted work in the previous 3 years, no other relationships or activities that could appear to have influenced the submitted work.

Ethics approval IRB00002556, Medical University Graz.

Provenance and peer review Not commissioned; externally peer reviewed.

REFERENCES

- 1 Singh AD, Topham A. Incidence of uveal melanoma in the United States: 1973–1997. *Ophthalmology* 2003;110:956–61.
- 2 Singh AD, Turell ME, Topham AK. Uveal melanoma: trends in incidence, treatment, and survival. *Ophthalmology* 2011;118:1881–5.

- 3 Singh AD, Topham A. Survival rates with uveal melanoma in the United States: 1973–1997. *Ophthalmology* 2003;110:962–5.
- 4 Lunsford LD, Flickinger J, Lindner G, *et al.* Stereotactic radiosurgery of the brain using the first United States 201 cobalt-60 source Gamma Knife. *Neurosurgery* 1989;24:151–9.
- 5 Marchini G, Gerosa M, Piovon E, *et al.* Gamma Knife stereotactic radiosurgery for uveal melanoma: clinical results after 2 years. *Stereotact Funct Neurosurg* 1996;66 (Suppl 1):208–13.
- 6 Rennie I, Forster D, Kemeny A, *et al.* The use of single fraction Leksell stereotactic radiosurgery in the treatment of uveal melanoma. *Acta Ophthalmol Scand* 1996;74:558–62.
- 7 Langmann G, Pendl G, Klaus M, *et al.* Gamma Knife radiosurgery for uveal melanomas: an 8-year experience. *J Neurosurg* 2000;93(Suppl 3):184–8.
- 8 Zehetmayer M, Menapace R, Kitz K, *et al.* Stereotactic irradiation of uveal melanoma with the Leksell gamma unit. *Front Radiat Ther Oncol* 1997;30:47–55.
- 9 Mueller AJ, Talies S, Schaller UC, *et al.* Stereotactic radiosurgery of large uveal melanomas with the Gamma-Knife. *Ophthalmology* 2000;107:1381–7;discussion 7–8.
- 10 Georgopoulos M, Zehetmayer M, Ruhschworm I, *et al.* Tumour regression of uveal melanoma after ruthenium-106 brachytherapy or stereotactic radiotherapy with Gamma Knife or linear accelerator. *Ophthalmologica* 2003;217:315–19.
- 11 Modorati G, Miserocchi E, Galli L, *et al.* Gamma Knife radiosurgery for uveal melanoma: 12 years of experience. *Br J Ophthalmol* 2009;93:40–4.
- 12 Song WK, Yang WL, Byeon SH, *et al.* Clinicopathologic report of uveal melanoma with persistent exudative retinal detachment after Gamma Knife radiosurgery. *Ophthalmologica* 2010;224:16–21.
- 13 Toktas ZO, Bicer A, Demirci G, *et al.* Gamma Knife stereotactic radiosurgery yields good long-term outcomes for low-volume uveal melanomas without intraocular complications. *J Clin Neurosci* 2010;17:441–5.
- 14 Mullner K, Langmann G, Pendl G, *et al.* Echographic findings in uveal melanomas treated with the Leksell Gamma Knife. *Br J Ophthalmol* 1998;82:154–8.
- 15 Langmann G, Wackernagel W, Stucklschweiger G, *et al.* Dose-volume histogram regression analysis of uveal melanomas after single fraction Gamma Knife radiosurgery. *Ophthalmologie* 2004;101:1111–19.
- 16 Haas A, Pinter O, Papaefthymiou G, *et al.* Incidence of radiation retinopathy after high-dosage single-fraction Gamma Knife radiosurgery for choroidal melanoma. *Ophthalmology* 2002;109:909–13.
- 17 Langmann G, Pendl G, Mullner K, *et al.* High-compared with low-dose radiosurgery for uveal melanomas. *J Neurosurg* 2002;97:640–3.
- 18 Damato B, Patel I, Campbell IR, *et al.* Visual acuity after Ruthenium(106) brachytherapy of choroidal melanomas. *Int J Radiat Oncol Biol Phys* 2005;63:392–400.
- 19 Seddon JM, Gragoudas ES, Polivogianis L, *et al.* Visual outcome after proton beam irradiation of uveal melanoma. *Ophthalmology* 1986;93:666–74.
- 20 Gragoudas E, Li W, Goitein M, *et al.* Evidence-based estimates of outcome in patients irradiated for intraocular melanoma. *Arch Ophthalmol* 2002;120:1665–71.
- 21 Damato B, Kacperek A, Chopra M, *et al.* Proton beam radiotherapy of choroidal melanoma: the liverpool-clatterbridge experience. *Int J Radiat Oncol Biol Phys* 2005;62:1405–11.
- 22 Kivela T, Eskelin S, Makitie T, *et al.* Exudative retinal detachment from malignant uveal melanoma: predictors and prognostic significance. *Invest Ophthalmol Vis Sci* 2001;42:2085–93.
- 23 Dunavoelgyi R, Dieckmann K, Gleiss A, *et al.* Local tumor control, visual acuity, and survival after hypofractionated stereotactic photon radiotherapy of choroidal melanoma in 212 patients treated between 1997 and 2007. *Int J Radiat Oncol Biol Phys* 2011;81:199–205.
- 24 Conway RM, Poothullil AM, Daftari IK, *et al.* Estimates of ocular and visual retention following treatment of extra-large uveal melanomas by proton beam radiotherapy. *Arch Ophthalmol* 2006;124:838–43.
- 25 Logani S, Cho AS, Ali BH, *et al.* Single-dose compared with fractionated-dose radiation of the OM431 choroidal melanoma cell line. *Am J Ophthalmol* 1995;120:506–10.